Open Source CFD

Tutorial #5: 3D Y-Pipe with FreeCAD

Incompressible Steady Flow: 3D Y-Pipe with simpleFOAM in FreeCAD

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Learning Objectives:

- Set up a complete CFD case using FreeCAD and CfdOF workbench
- Define boundary conditions visually in the GUI
- Generate mesh automatically with snappyHexMesh
- Run simpleFoam directly from FreeCAD
- Post-process results using ParaView integration
- Compare GUI vs terminal workflows (Tutorial #4)

This tutorial shows the work flow for the simulation of a 3D Y-shaped pipe geometry with FreeCAD and the CfdOf workbench. The geometry generated in Tutorial #4 will be reused. Instead of editing manually the dictionaries, in this case all will be done with the GUI. It is more efficient and reduces the risk of typographic mistakes. On the other side, a graphical interface, that often is not available in large computers, is needed.

Contents

1 Opening the Geometry in FreeCAD

1. **Launch FreeCAD** from the terminal or applications menu:
2. **Open the Y-pipe geometry**:
   - Go to `File` → `Open`
   - Navigate to your Tutorial #4 folder
   - Select `Ypipe.FCStd`
   - Click `Open`

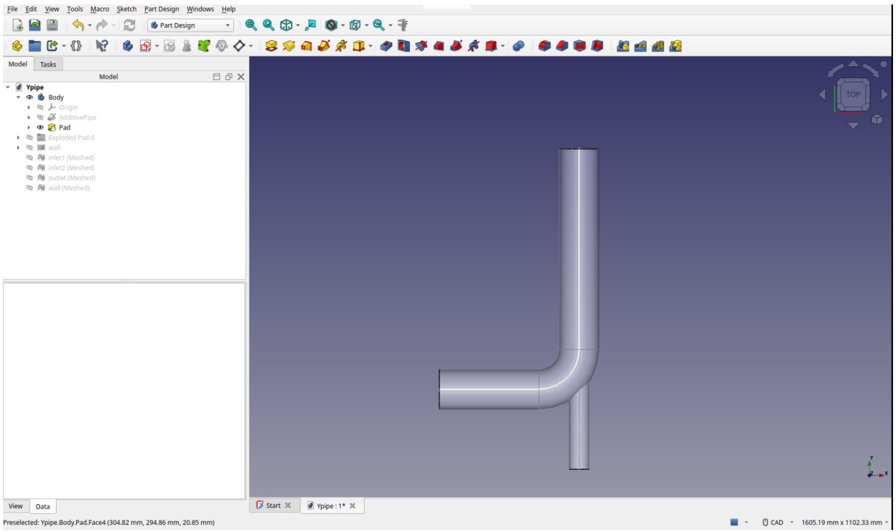

Figure 1: Opening the Y-pipe geometry

3. **Verify the geometry**:
   - The model should appear in the 3D view
   - The tree view should show the `Body` with all features
   - Rotate and zoom to inspect all parts

   TIP: If you don't have the geometry file, follow the geometry creation steps from Tutorial #4, Sections 2.1-2.2.

2 Setting Up the CFD Analysis

2.1 Activating CfdOF Workbench

1. **Switch to CfdOF workbench**:
    - From the workbench dropdown, select **CfdOF**
2. **Change output directory**
    - Select the menu `CfdOf` → `Open preferences`
    - Change the `default output directory` to your folder for the Tutorial #5.

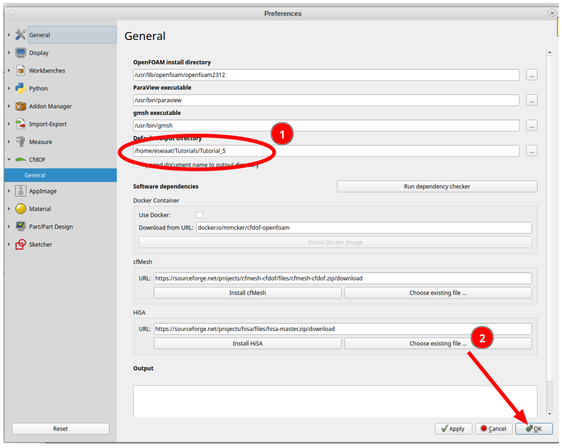

Figure 2: Selecting of output directory

3. **Create a new analysis**:
    - Click the **Analysis container** button in the toolbar
    - A new `CfdAnalysis` object appears in the tree

2.2 Defining the Simulation

1. **Set physics models**:
    - Click on `Select models` button
    - In the Tasks panel, set:
        - `Flow type`: `Incompressible`
        - `Time`: `Steady`
        - `Turbulence model`: `RANS k-omega SST`

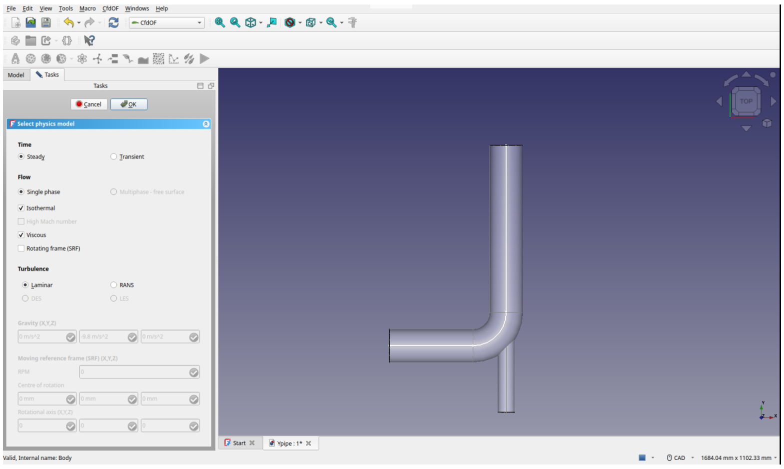

Figure 3: Physics models

2. **Select fluid material**:
    - Click **Add fluid properties** button
    - Choose `Water` from the library
    - Verify properties: $\rho = 1000\,\mathrm{kg/m^3}$, $\nu = 10^{-3}\,\mathrm{Pa \cdot s}$

3 Defining Boundary Conditions

3.1 Identifying Boundary Faces

The Y-pipe has four distinct boundary regions:

- **inlet1**: Secondary inlet (horizontal branch)
- **inlet2**: Primary inlet (vertical pipe)
- **outlet**: Pipe outlet
- **wall**: All remaining surfaces

3.2 Creating Boundary Conditions

1. **Add inlet2**:
    - Select `inlet2` face (secondary inlet). It will become green.
    - Click **Fluid boundary** button. It will become blue
    - `Type: Inlet, Uniform velocity`

- `Velocity`: `(0 4 0)` m/s (vertical direction)
- `Pressure`: `0` Pa (relative pressure)
- Click `OK`
- `Turbulence specification: Intensity & Length Scale`
  - `Intensity (I)`: 5%
  - `Length Scale (l)`: 1.4 mm
- Rename it as `inletSecondary`

TIP: `CfdOf` automatically calculates k and ω from these parameters using standard formulas: $k = \frac{3}{2}(UI)^2$, $\omega = \frac{\sqrt{k}}{C_\mu^{1/4} L}$ where $C_\mu = 0.09$ and L is the mixing length scale.

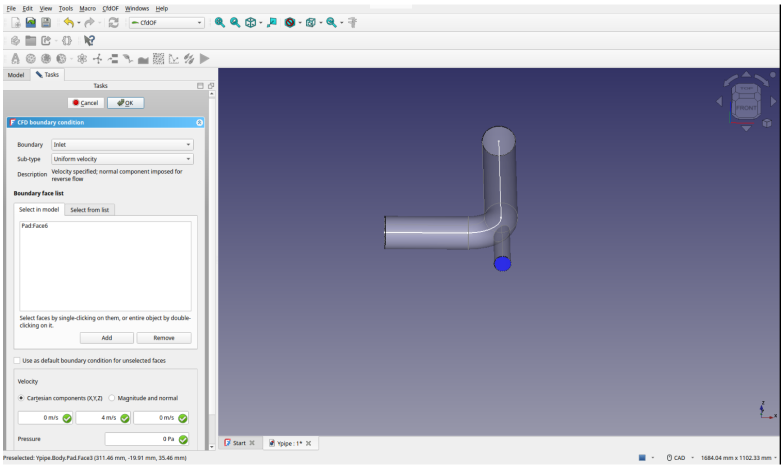

Figure 4: Setting inlet boundary conditions

2. **Add inlet1**:
   - Repeat the same steps as with `inlet2`
   - `Type`: `inlet, Total pressure`
   - Set `Pressure` : 0 Pa
   - Same turbulence specification as in `inlet2`
   - Rename it as `inletMain`
3. **Add outlet**:
   - Repeat the same steps
   - `Type`: `outlet, Static pressure`

- `Pressure: 0 Pa` (static)
- Leave the default name (`outlet` is not allowed because there is already an element in the tree with this name)

4. **Add walls**:
   - Select all remaining faces (use Ctrl+click for multiple selection)
   - `Type: wall`
   - `Name: walls`
5. **Verify all boundaries**:
   - The tree should show four boundary conditions
   - Each should have a distinct color in the 3D view

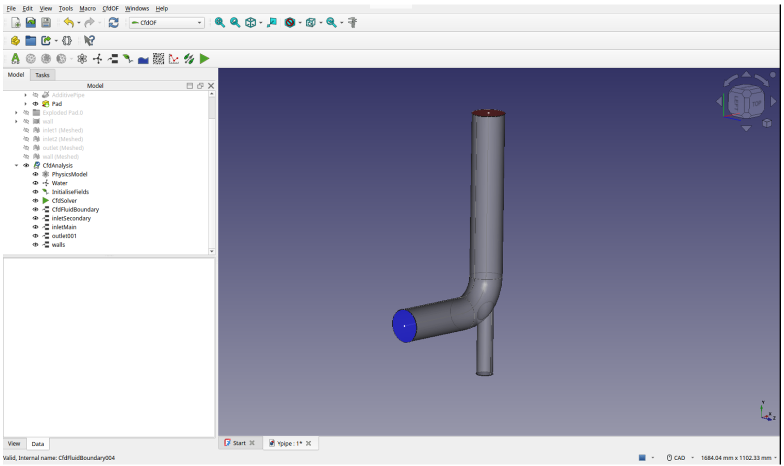

Figure 5: Completed boundary conditions